

Classical Mechanics 1 at CMI: Mid-semester exam, 27 Sept 2025

This question paper has 1 page. Read questions carefully, write your answers clearly in blue and briefly explain your reasoning. Start each problem on a new sheet of paper. No computers/phones/notes/books/discussion permitted: closed book exam. 60 marks. 3 hrs.

1. $\langle 6 \rangle$ Suppose we make a linear change from Cartesian coordinates (x, y) to new coordinates (u, v) on the plane, given by $u = ax + by$ and $v = cx + dy$ for some real constants a, b, c, d . It is possible to show (you are not asked to do this) that vectors pointing in the direction of increasing u and v holding the other fixed are given by $\mathbf{u} = a\hat{x} + b\hat{y}$ and $\mathbf{v} = c\hat{x} + d\hat{y}$ where $\hat{x}$ and $\hat{y}$ are the usual unit vectors in the directions of increasing x and y respectively. Find conditions on (a, b, c, d) to guarantee that $\mathbf{u}$ and $\mathbf{v}$ are orthonormal. Interpret the answer in terms of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$.
2. $\langle 3 + 3 + 4 \rangle$ (a) Draw a figure indicating Cartesian and spherical polar coordinates in 3d Euclidean space with a chosen origin O . You must indicate O , the Cartesian axes and the coordinates (x, y, z) and (r, θ, ϕ) of a point P with position vector $\mathbf{r}$. (b) Give formulae for x, y, z in terms of r, θ, ϕ . Give names for r, θ, ϕ . (c) Suppose $(r(t), \theta(t), \phi(t))$ are the coordinates of a particle at time t . Without detailed calculations, what can you say about the vector $\frac{d\hat{\theta}(t)}{dt}$ where $\hat{\theta}(t)$ is the unit vector in the appropriate direction at the point on the trajectory with time label t .
3. $\langle 8 \rangle$ In an explosion, a spherically symmetric expanding blast wave is produced. It may be argued that to first approximation, the radius of this blast wave R can only depend on the total energy E of the explosion, the ambient mass density ρ of the gas into which the blast wave propagates and on time. Use dimensional analysis to deduce an expression for $R(t)$ including any dimensionless constants.
4. $\langle 3 + 3 + 4 \rangle$ Consider the force field on the x - y plane $\mathbf{f} = y\hat{x}$. (a) Plot this vector field using arrows to indicate the size and direction of the vectors at various points on the plane. (b) Try to express $\mathbf{f}$ as the negative gradient of a potential function ϕ and explain what you find. (c) Is $\mathbf{f}$ a conservative force field? Illustrate your answer by calculating the work done by this force along two sample (say, piecewise straight line) paths γ_1 and γ_2 between $P(0, 0)$ and $Q(1, 0)$.
5. $\langle 4 + 4 \rangle$ (a) State Newton's first law of mechanics. (b) Give an example of a frame that may be considered inertial to a reasonable approximation for a suitable (to be specified through some examples) class of physical phenomena and one that is not inertial.
6. $\langle 3 + 4 + 2 + 3 + 4 + 2 \rangle$ Consider a particle of mass m that moves along a line subject to the restoring force $-kx\hat{x}$ where $x(t)$ is the displacement from its equilibrium position. (a) Write Newton's equation of motion for $x(t)$ and mention the independent and dependent variables. (b) What sort of equation is it? Mention suitable attributes. Is the equation linear or nonlinear? (c) Give a formula and name for one conserved quantity, verify it is conserved. (d) Write a formula for a solution $x(t)$ of the equation of motion with oscillation amplitude A and initial displacement $A \cos \phi$. Check that your $x(t)$ is a trajectory. (e) What is the phase space of this system? Sketch a phase portrait showing a few trajectories (both static and nonstatic). (f) Explain the direction of motion along the trajectories on phase space by considering a sample nonstatic trajectory.